

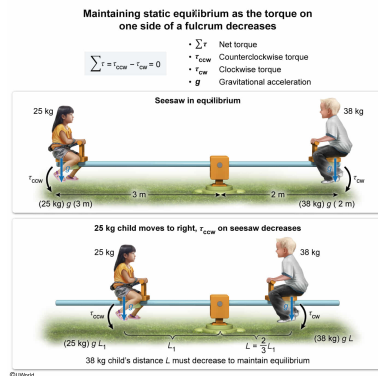
A seesaw is in equilibrium when a 25-kg child sits 3 m to the left of the fulcrum and a 38-kg child sits 2 m to the right of the fulcrum. If the 25-kg child moves to the right, how must the 38-kg child's distance L from the fulcrum change for the seesaw to remain in equilibrium?

- ✓ ☒ A. L must decrease, because the torque produced by the 25-kg child decreases. [74%]
☐ B. L must not change, because each child's weight is constant. [1%]
☐ C. L must not change, because the torque produced by the 25-kg child is constant. [2%]
☐ D. L must increase, because the torque produced by the 25-kg increases. [21%]

Correct

74% answered correctly

Explanation:



Torque τ is a rotational force that causes an object to spin about an axis. By convention, torques that cause an object to spin counterclockwise are positive, and torques that spin an object clockwise are negative. When an object is in **static equilibrium** the **net torque** (ie, the sum of all the torques exerted on the object) is zero:

$$\sum \tau = 0$$

In this question, the **weights** of each child are exerted downward, perpendicular to the seesaw, which is in static equilibrium. Consequently, the torques due to each child, which are the product of each child's mass, the gravitational acceleration g , and their respective distance from the fulcrum, sum to zero:

$$\sum \tau = (25 \text{ kg})g(3 \text{ m}) - (38 \text{ kg})g(2 \text{ m}) = 0$$

In general, when the 25-kg child is at distance L_1 from the fulcrum and the 38-kg child is at distance L from the fulcrum, the static equilibrium condition implies that:

$$\sum \tau = (25 \text{ kg}) g L_1 - (38 \text{ kg}) g L = 0$$

As the 25-kg child moves to the right, L_1 *decreases* and the corresponding torque decreases. Solving for L of the 38-kg child yields:

$$(38 \text{ kg})gL = (25 \text{ kg})gL_1$$

$$L = \frac{25}{38}L_1 \approx \frac{2}{3}L_1$$

Hence, L is **directly proportional** to L_1 . Therefore, L must decrease because the torque due to the 25-kg child decreases.

(Choice B) Each child's weight is constant, but L must decrease as the 25-kg child moves to the right.

(Choice C) Torque depends on the distance from the fulcrum. Hence, the torque decreases as the child's distance from the fulcrum decreases.

(Choice D) As the 25-kg child moves to the right, the torque produced on the seesaw will decrease, not increase.

Educational objective:

The net torque exerted on an object in static equilibrium is zero. If the torque on one side of a fulcrum decreases, the torque on the other side of the fulcrum must also decrease for the system to remain in equilibrium.

Time spent: 0 sec
 Subject:
 Physics

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 System:
 Mechanics and Energy